

Playwright Automation Framework – Complete Quick Reference

This redesigned guide provides a structured and modern reference for the Playwright Automation Framework, covering setup, execution, locators, waits, assertions, API mocking, authentication, and debugging.

1. Installation & Setup

```
npm init playwright@latest
npm install -D @playwright/test
npx playwright install
npx playwright install chromium firefox webkit
```

2. Test Execution Commands

```
npx playwright test
npx playwright test --headed
npx playwright test --ui
npx playwright test --debug
npx playwright test --project=chromium
npx playwright test tests/test.spec.ts
```

3. Browser & Context Management

```
const browser = await chromium.launch();
const browserHeaded = await chromium.launch({ headless: false });
await browser.close();

const context = await browser.newContext();
const authContext = await browser.newContext({ storageState: 'state.json' });
await context.storageState({ path: 'state.json' });
```

4. Page Handling & Navigation

```
const page = await context.newPage();
await page.goto('https://example.com');
await page.reload();
await page.goBack();
await page.goForward();
await page.close();
```

5. Multiple Tabs / Windows

```
const [newPage] = await Promise.all([
  context.waitForEvent('page'),
  page.click('a[target=_blank]')
]);
```

6. Locators (All Types)

```
page.locator('css-selector');
page.locator('//xpath-selector');
page.getByText('Login');
page.getByRole('button', { name: 'Submit' });
page.getByLabel('Email');
page.getByPlaceholder('Enter Search Query');
page.getByTestId('submit-btn');
```

7. User Interactions

```
await locator.click();
await locator.fill('Standard text input');
await locator.type('Slow typing simulation');
await locator.clear();
await locator.check();
await locator.uncheck();
await locator.selectOption('Value_1');
```

8. Advanced Mouse & Keyboard

```
await page.dblclick('selector');
await page.hover('selector');
await page.dragAndDrop('#source', '#target');
await page.mouse.move(100, 200);

await page.press('#input', 'Enter');
await page.keyboard.type('Hello World');
await page.keyboard.press('Control+A');
await page.keyboard.press('Backspace');
```

9. Waiting & Synchronization

```
await page.waitForSelector('.loading-complete');
await page.waitForLoadState('networkidle');
await page.waitForURL('**/dashboard');
await page.waitForTimeout(3000);
```

10. Assertions (Expect)

```
await expect(page).toHaveURL('https://site.com/home');
await expect(page).toHaveTitle('Welcome');
await expect(locator).toBeVisible();
await expect(locator).toBeHidden();
await expect(locator).toHaveText('Success');
await expect(locator).toContainText('Part of string');
await expect(locator).toBeEnabled();
```

11. Frames, iFrames & Dialogs

```
const frame = page.frame({ name: 'mainFrame' });
await frame.locator('#btn').click();

page.on('dialog', dialog => dialog.accept());
```

```
page.on('dialog', dialog => dialog.dismiss());
```

12. File Operations (Upload & Download)

```
await page.setInputFiles('#upload', 'resume.pdf');

const downloadPromise = page.waitForEvent('download');
await page.getByText('Download File').click();
const download = await downloadPromise;
await download.saveAs('local_file.pdf');
```

13. Authentication & Storage

```
await context.addCookies([{ name: 'session', value: '123' }]);
const cookies = await context.cookies();
await context.clearCookies();
await context.storageState({ path: 'auth.json' });
```

14. API Testing & Mocking

```
const response = await request.get('/api/v1/users');
expect(response.status()).toBe(200);

await page.route('**/api/**', route => {
  route.fulfill({ status: 200, body: JSON.stringify({ data: 'mocked' }) });
});

await page.route('**/*.png,jpg,jpeg', route => route.abort());
```

15. Test Hooks & Configuration

```
test.beforeAll(async () => {});
test.beforeEach(async ({ page }) => {});
test.afterEach(async () => {});

test.describe.configure({ mode: 'parallel', retries: 2 });
```

16. Debugging & Reporting

```
npx playwright test --trace on
npx playwright show-trace trace.zip
npx playwright show-report
```

17. Pro Interview Commands

```
test.only();
test.skip();
test.fail();
test.slow();
ENV=qa npx playwright test
```